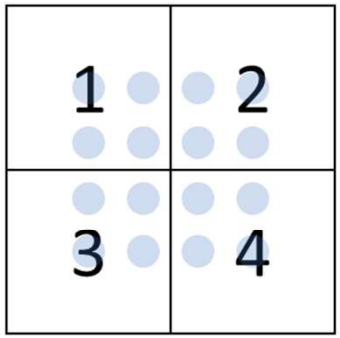
Souece: https://en.wikipedia.org/wiki/Bilinear_interpolation

In-Network Upsampling: Bilinear Interpolation



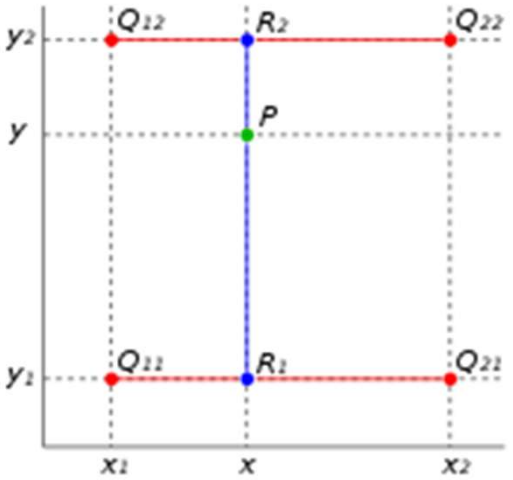
1.00	1.25	1.75	2.00
1.50	1.75	2.25	2.50
2.50	2.75	3.25	3.50
3.00	3.25	3.75	4.00

Input: C x 2 x 2

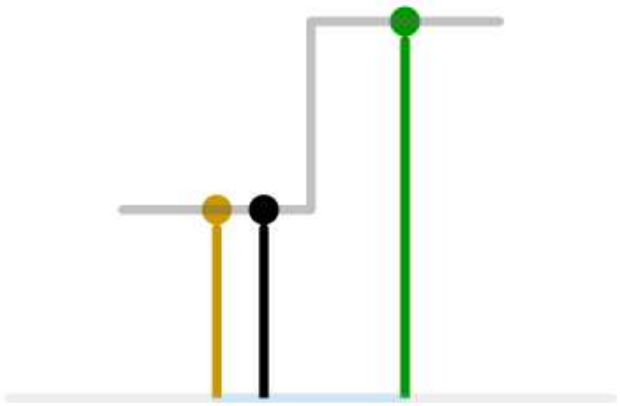
Output: C x 4 x 4

$$f(x, y_1) = \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21}),$$
$$f(x, y_2) = \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22}).$$

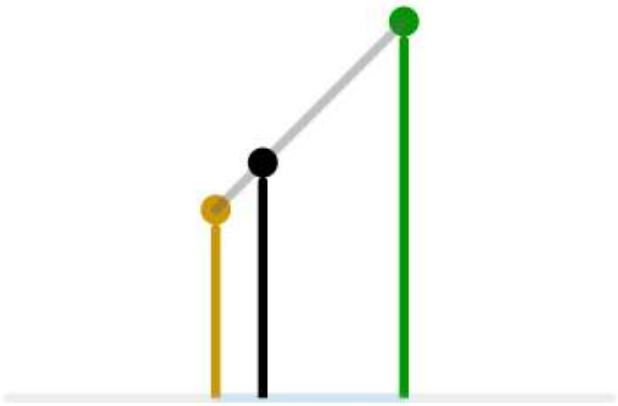
$$f(x, y) = \frac{y_2 - y}{y_2 - y_1} f(x, y_1) + \frac{y - y_1}{y_2 - y_1} f(x, y_2)$$



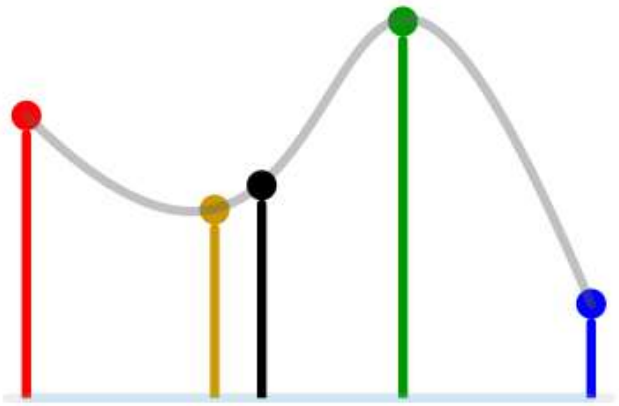
$Q_{11} = (x_1, y_1)$, $Q_{12} = (x_1, y_2)$,
$Q_{21} = (x_2, y_1)$, $Q_{22} = (x_2, y_2)$.



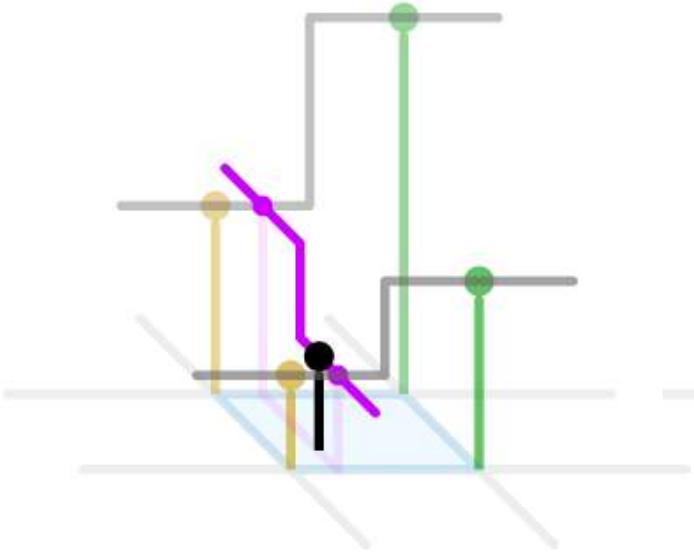
1D nearest-neighbour



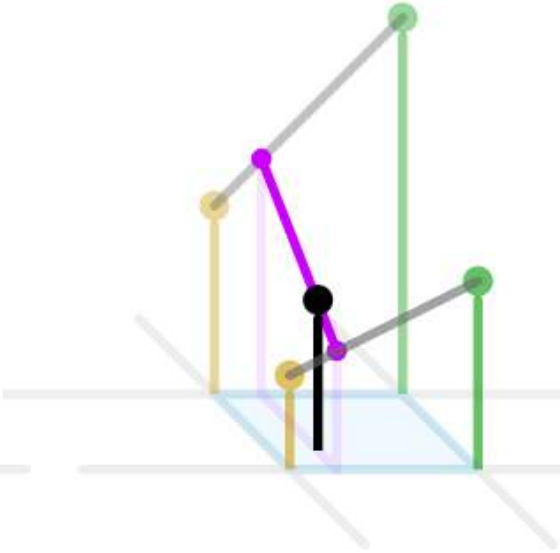
Linear



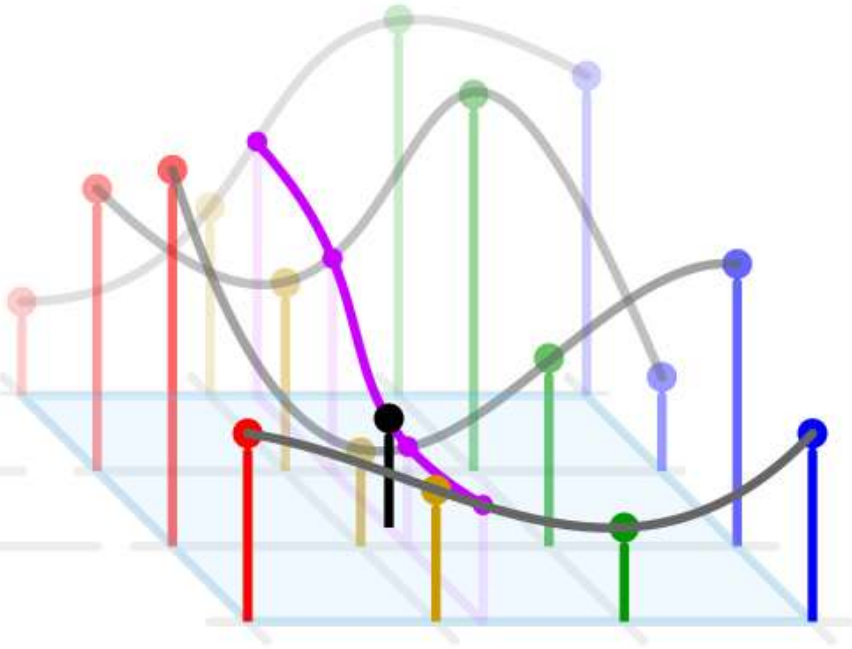
Cubic



2D nearest-neighbour



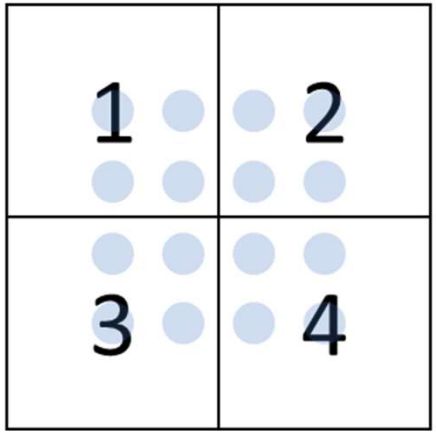
Bilinear



Bicubic

Source: https://en.wikipedia.org/wiki/Bicubic_interpolation

In-Network Upsampling: Bicubic Interpolation



0.68	1.02	1.56	1.89
1.35	1.68	2.23	2.56
2.44	2.77	3.32	3.65
3.11	3.44	3.98	4.32

Input: C x 2 x 2

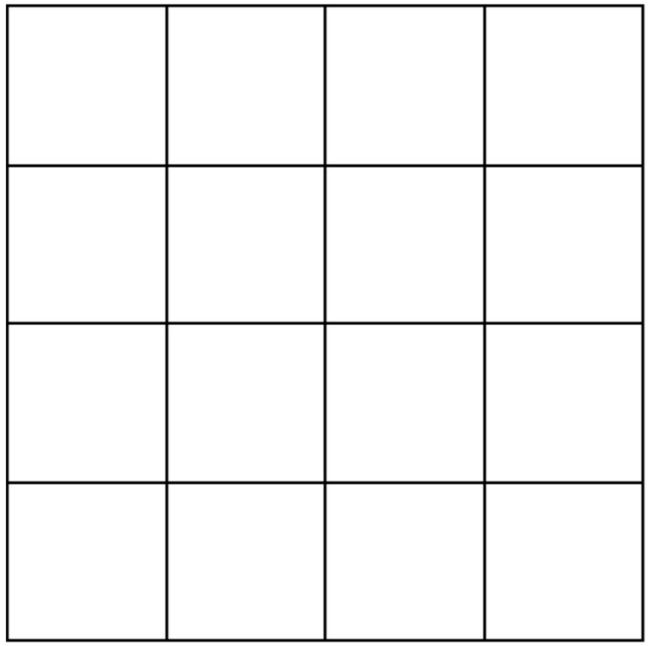
Output: C x 4 x 4

Szeliski, Chapter 6.3

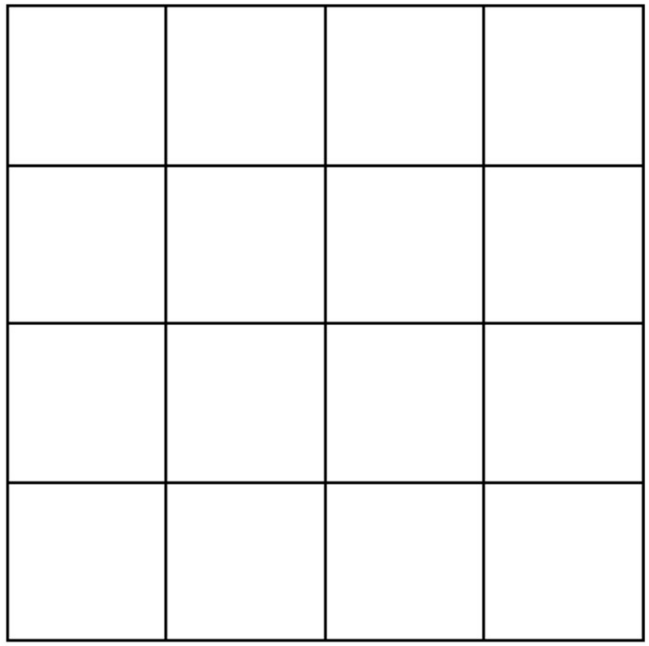
Upsampling: Transposed Convolution

➢Recall

- Normal 3 x 3 convolution, stride 1, pad 1



Input: 4 x 4

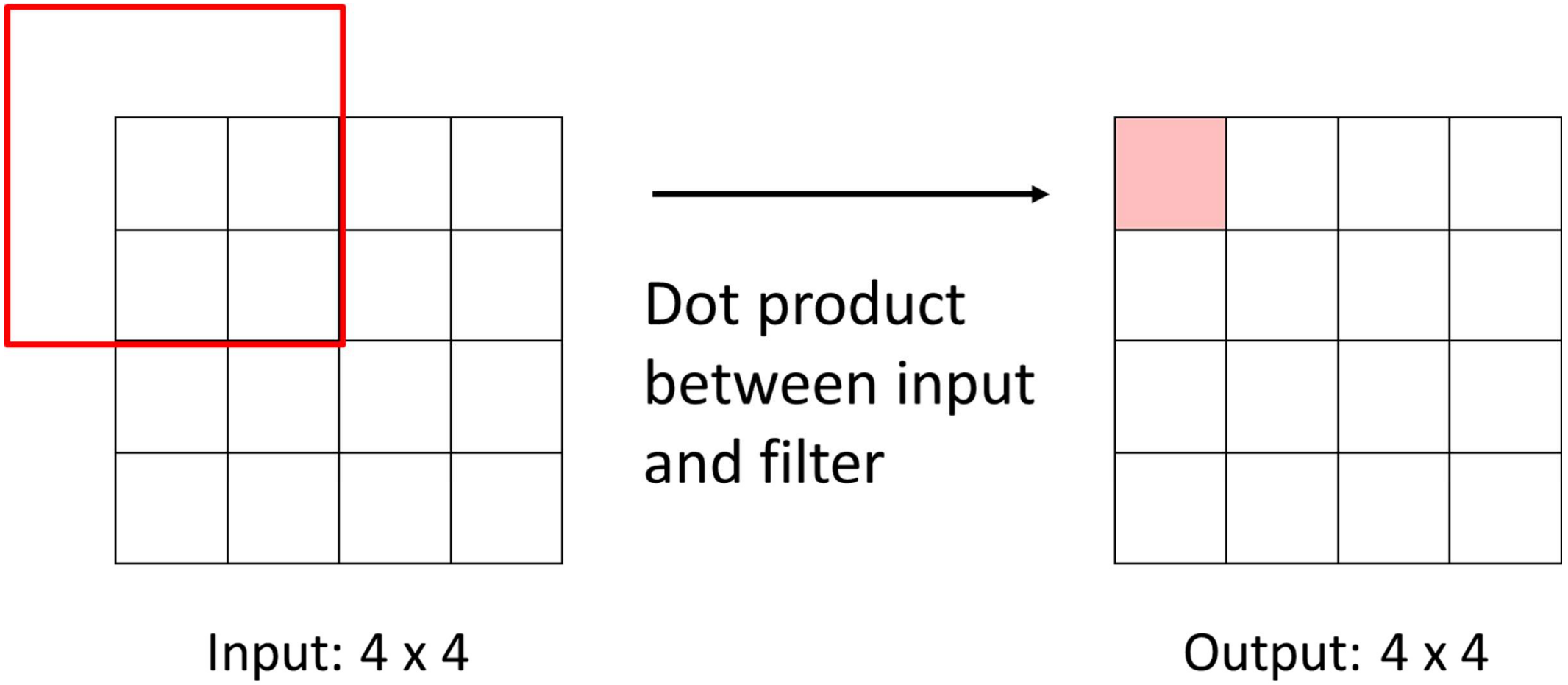


Output: 4 x 4

Szeliski, Chapter 6.3

Upsampling: Transposed Convolution

- **Recall**
  - Normal 3 x 3 convolution, stride 1, pad 1

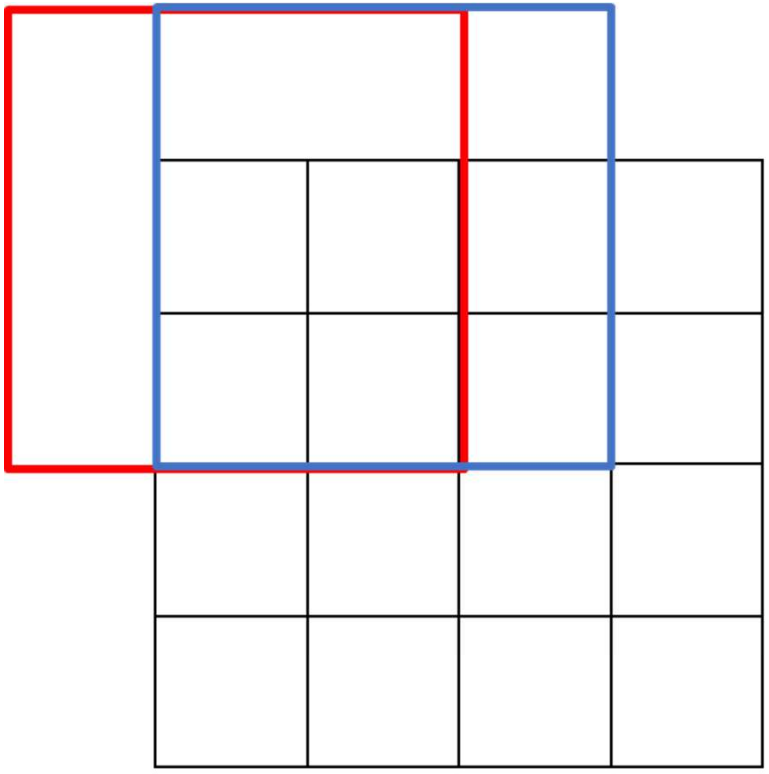


Szeliski, Chapter 6.3

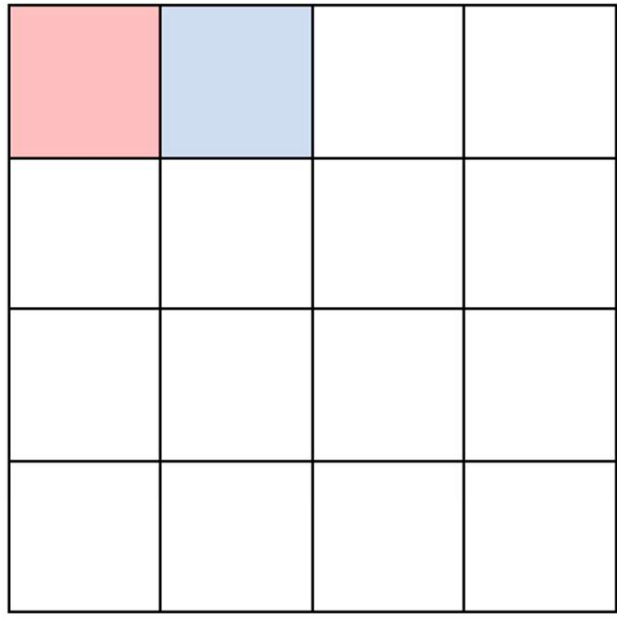
Upsampling: Transposed Convolution

➢ **Recall**

- Normal 3 x 3 convolution, stride 1, pad 1



Dot product between input and filter



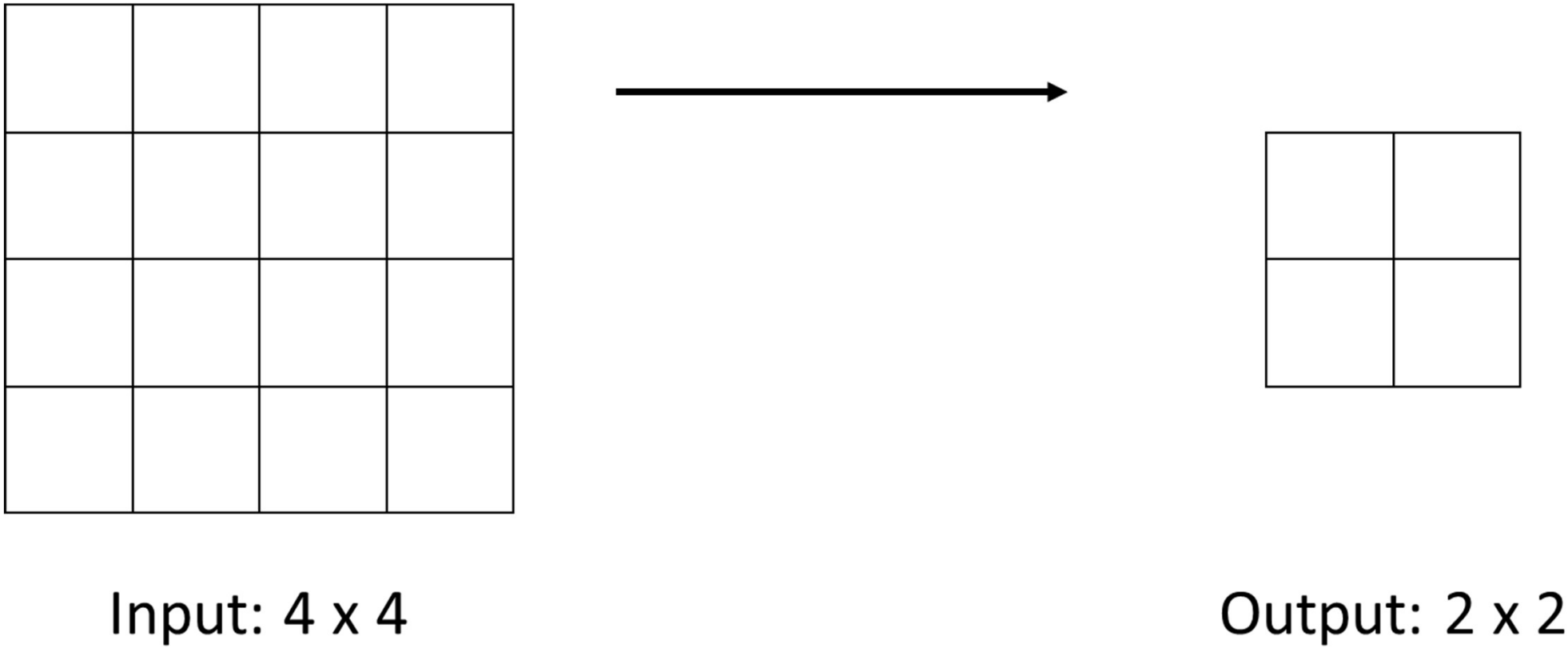
Input: 4 x 4

Output: 4 x 4

Szeliski, Chapter 6.3

Upsampling: Transposed Convolution

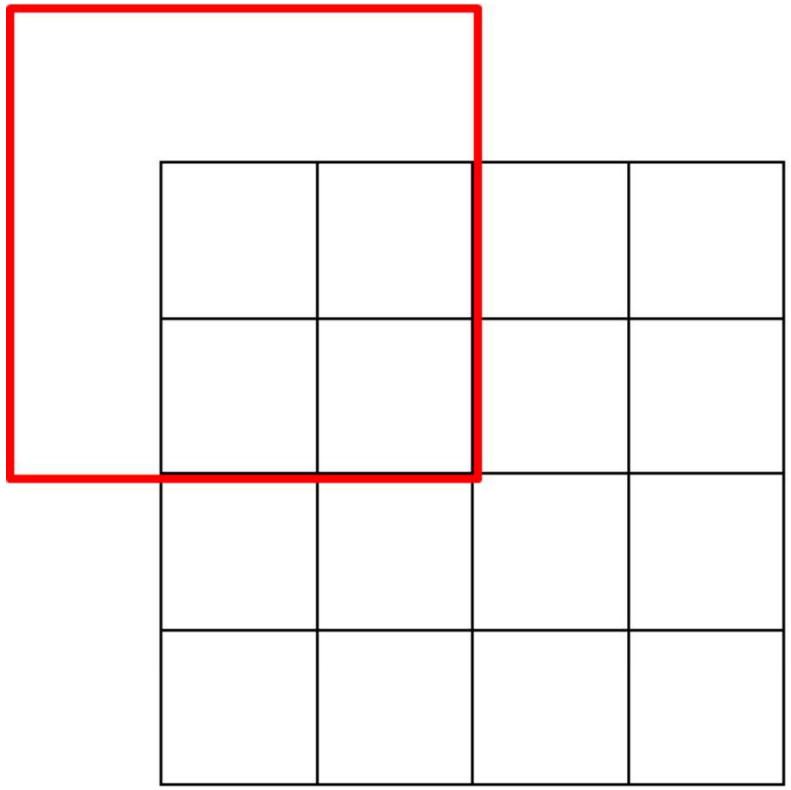
- **Recall**
  - Normal 3 x 3 convolution, stride 2, pad 1



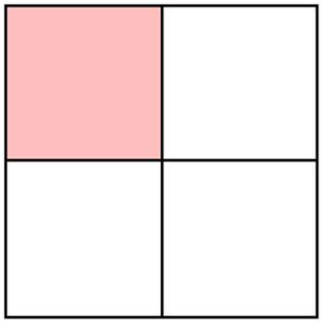
Szeliski, Chapter 6.3

Upsampling: Transposed Convolution

- **Recall**
  - Normal 3 x 3 convolution, stride 2, pad 1



Dot product between input and filter



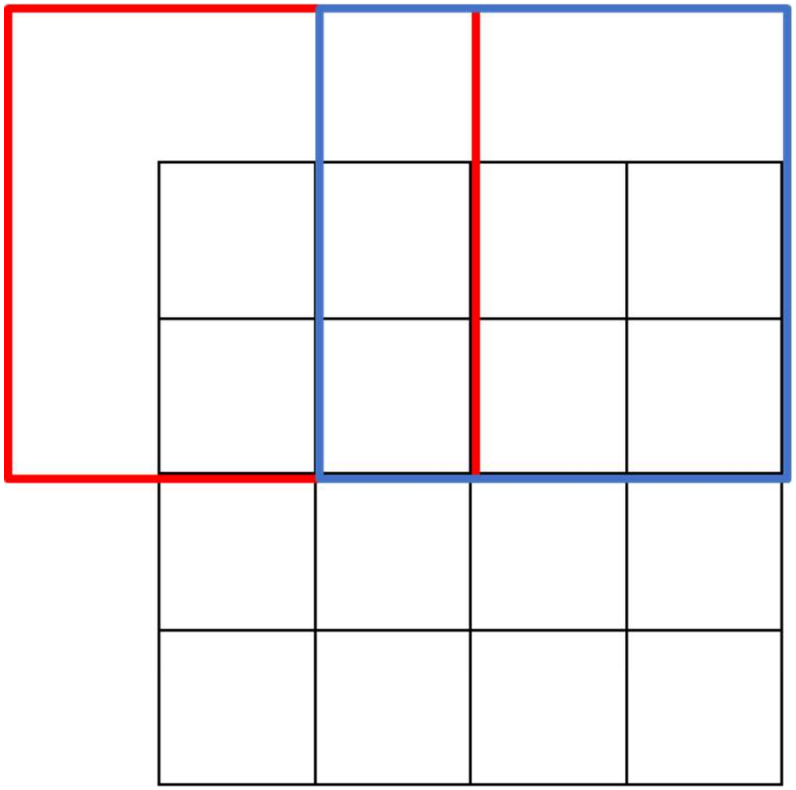
Input: 4 x 4

Output: 2 x 2

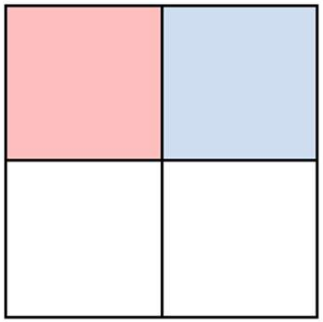
Szeliski, Chapter 6.3

Upsampling: Transposed Convolution

- **Recall**
  - Normal 3 x 3 convolution, stride 2, pad 1



Dot product between input and filter



Input: 4 x 4

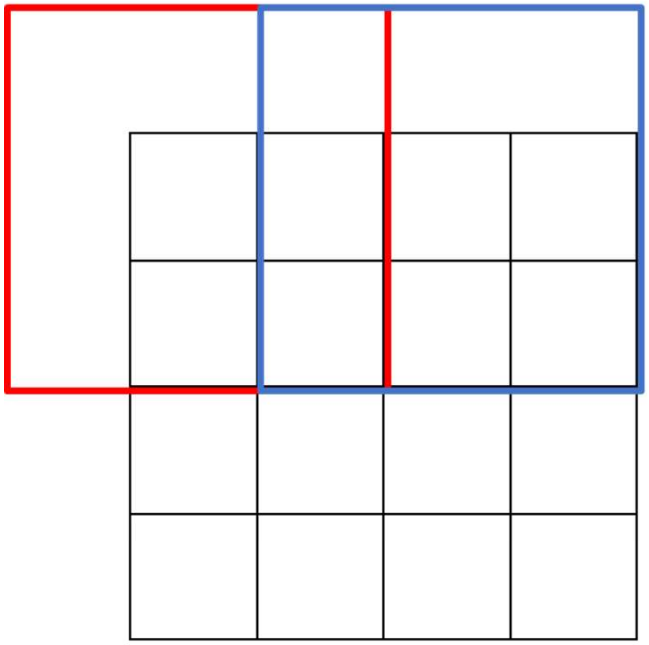
Output: 2 x 2

Szeliski, Chapter 6.3

Upsampling: Transposed Convolution

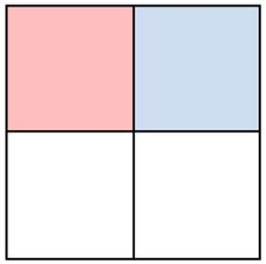
➢ Recall

- Normal 3 x 3 convolution, stride 2, pad 1



Convolution with stride > 1 is "Learnable Downsampling"
Can we use stride < 1 for "Learnable Upsampling"?

Dot product between input and filter



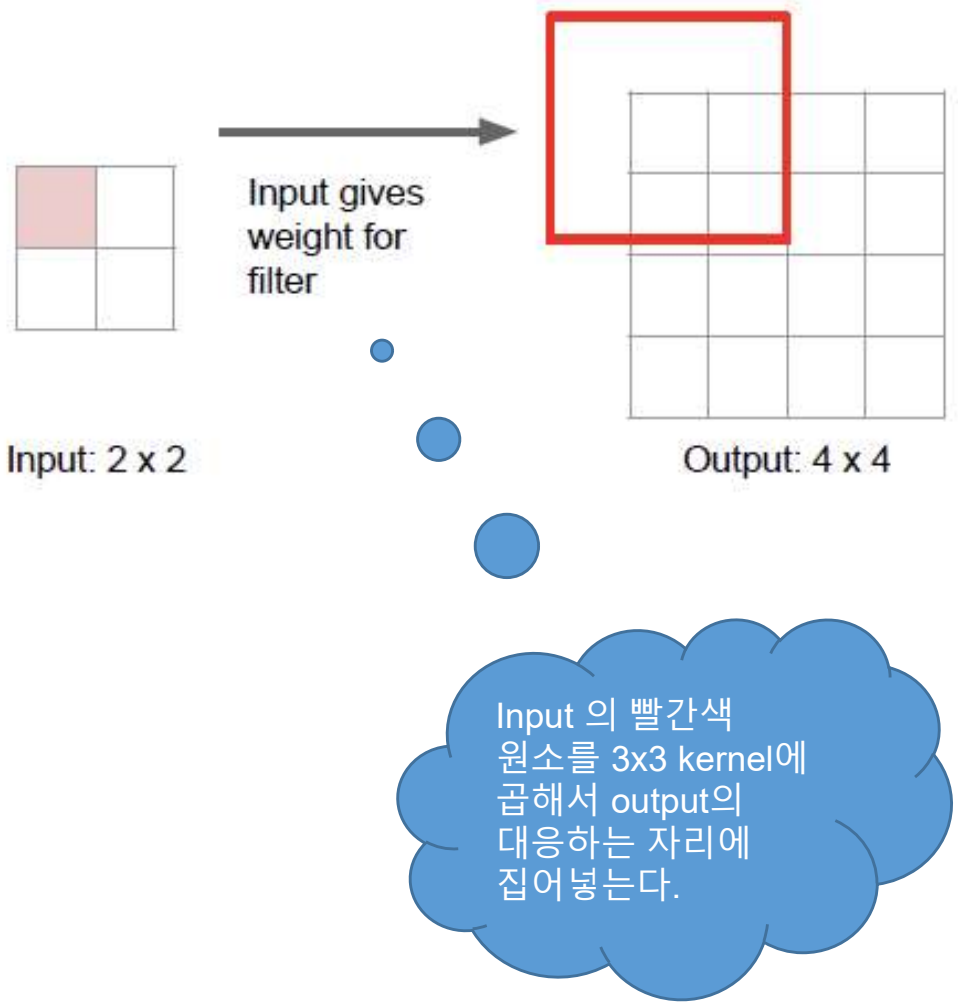
Input: 4 x 4

Output: 2 x 2

Szeliski, Chapter 6.3

Upsampling: Transposed Convolution

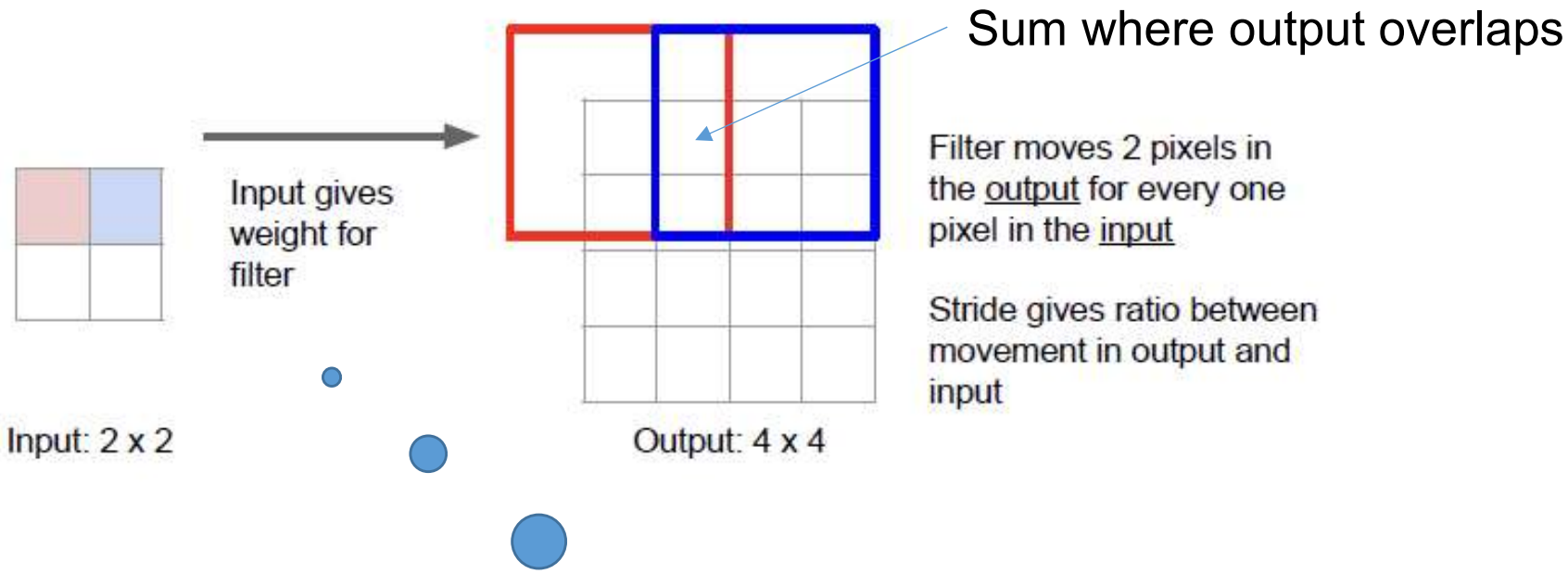
➢ 3 x 3 transposed convolution, stride 2, pad 1



Slide: cs231n, Stanford Univ. 2021

Upsampling: Transposed Convolution

➢ 3 x 3 transposed convolution, stride 2, pad 1



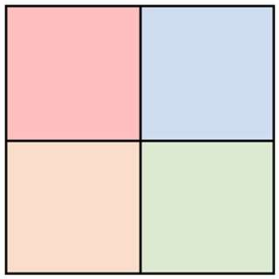
같은 방법으로 input의 파란색 원소를 3x3 kernel에 곱해서 output의 대응하는 위치에 집어넣는다.
이 때 output에 겹치는 구간이 발생하는데, 겹치는 부분의 값은 모두 더해준다

Slide: cs231n, Stanford Univ. 2021

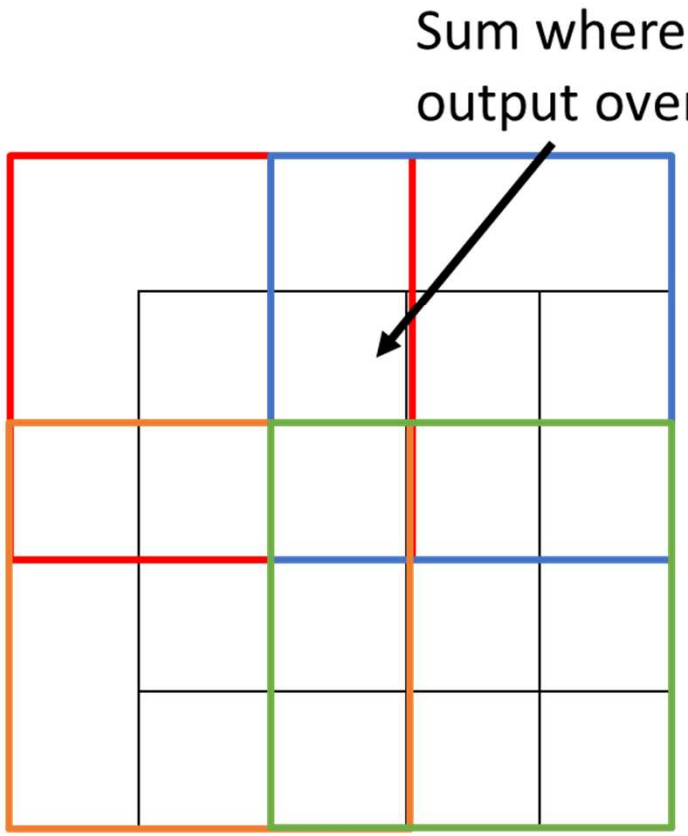
Upsampling: Transposed Convolution

➢ 3 x 3 transposed convolution, stride 2, pad 1

This gives 5x5 output – need to trim one pixel from top and left to give 4x4 output



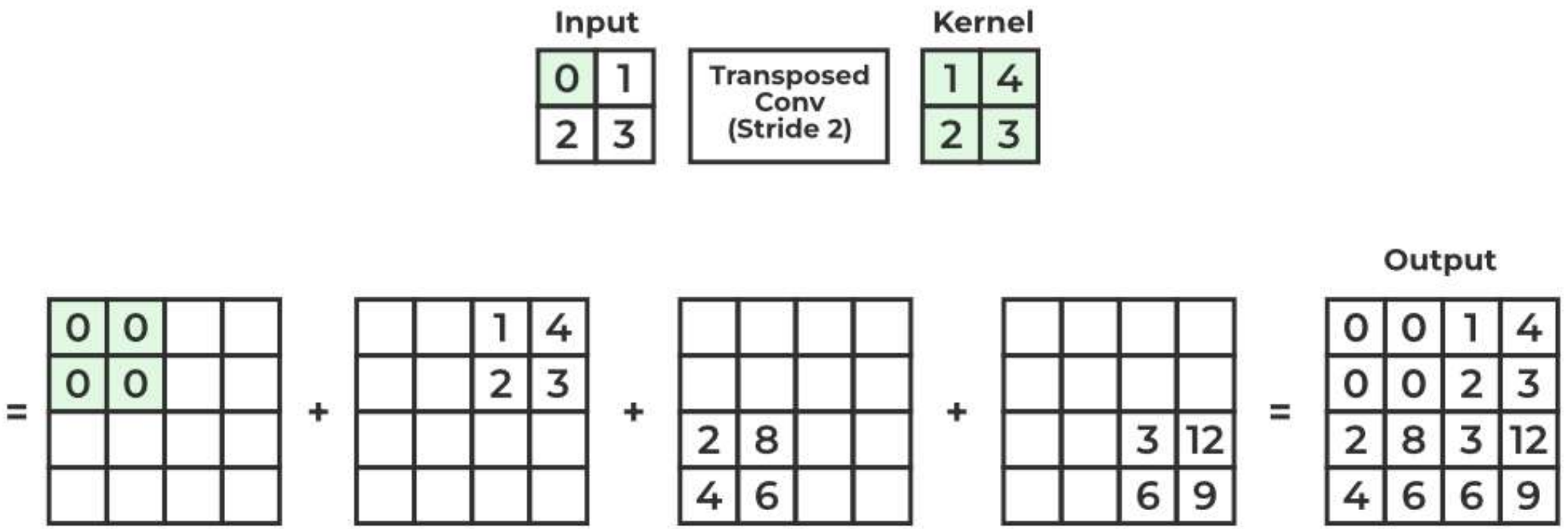
Weight filter by input value and copy to output



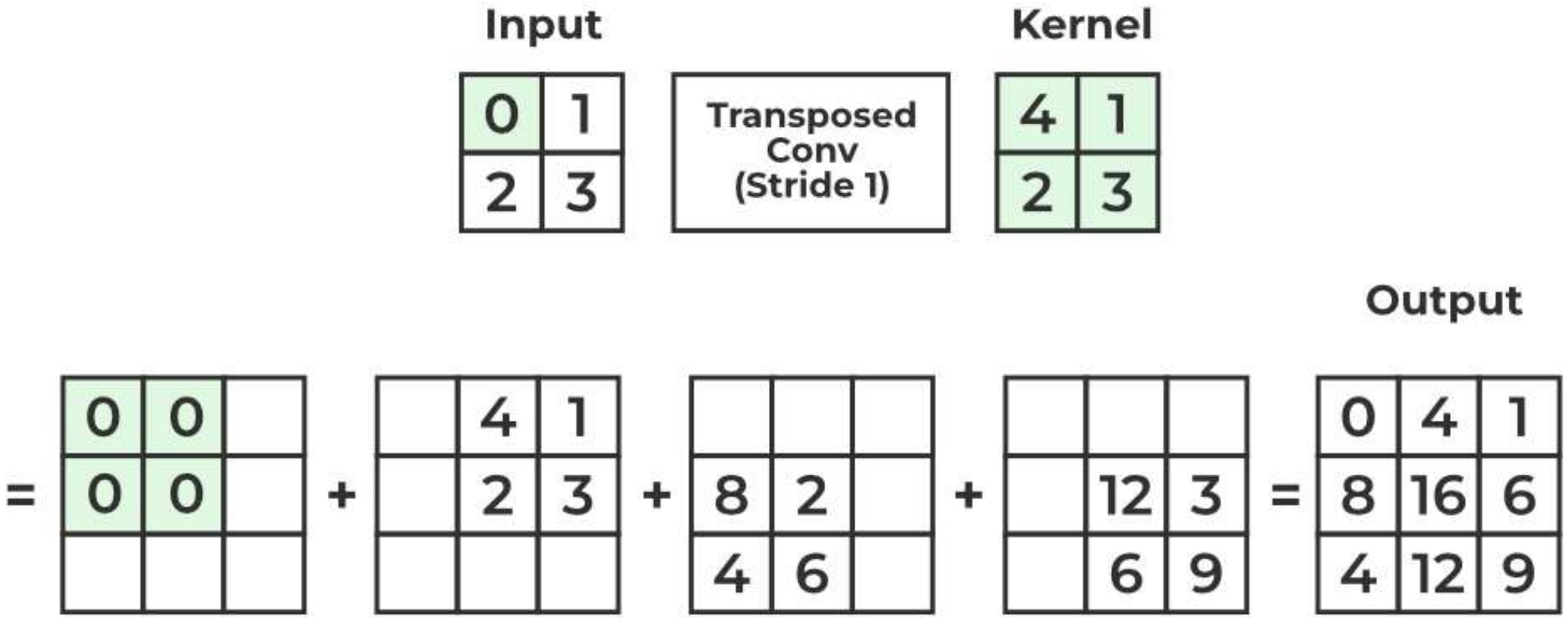
Input: 2 x 2

Output: 4 x 4

Slide: cs231n, Stanford Univ. 2021



Source: https://www.geeksforgeeks.org/what-is-transposed-convolutional-layer/



Source: https://www.geeksforgeeks.org/what-is-transposed-convolutional-layer/